TECHNICAL REPORT / 01

Hybrid Algorithm-Based 3D Container Loading Optimisation

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Question

Compare packing methods under common time limits and a repeatable attempt cap. Each method tries to fill one container with rectangular boxes while respecting rotation rules, avoiding overlap and staying within payload.

Main result

The three-seed comparison at 150 ms gave a hybrid-minus-First-Fit difference of +0.22 percentage points (paired 95% interval -0.68 to +1.07). The methods show different strengths across the three instance families.

Instance family	Evidence from the four common limits
Geometry	Results vary by limit, with some intervals spanning zero
Payload	Equal utilization at all four common limits
Mixed	Hybrid ahead at all four common limits

Under the four-attempt cap, the mean difference was +2.66 points (95% interval +0.95 to +4.51). Each First-Fit attempt constructs one layout. Each hybrid attempt builds a pool and searches it, so an equal attempt cap is a reproducible setting rather than a measure of equal computation.

What the follow-up measures

Thirty cases across geometry, payload and mixed families. Each score averages the three frozen trial seeds. The five-second check covers nine cases and is reported separately.

The follow-up reuses the same instances as the earlier held-out study and evaluates new search settings and a stronger baseline. Treat it as a follow-up on those instances.

Decision

Weight-aware First-Fit is a strong default for payload and geometry. The hybrid and exact pool selection are most useful to compare on mixed cases, where size and weight compete.

Methods and definitions

Common feasibility rules

Boxes must lie inside the container, use allowed orientations and appear at most once. Boxes may touch faces. Total weight must stay within payload. Utilization is selected volume divided by container volume. Optional full support requires every elevated base to be covered by boxes directly below.

Weight-aware multi-start First-Fit

Try decreasing volume first, then longest edge, volume per gram and a weight-only knapsack-guided order. Later restarts shuffle box order and allowed orientations. The solver keeps its highest-volume valid packing. The separate weight-only control solves the knapsack and packs only that subset.

Hybrid search

Build up to 160 placements by combining independent layouts. The follow-up varies the order of coordinate priorities used to pick corners. An attempt uses 45 sweeps and four reads. A compound move inserts a placement and removes its selected blockers. The visited states are conflict-free. The production search evaluates the payload penalty after minimizing binary slack analytically, so it searches a reduced objective.

Explicit QUBO and HiGHS

The expanded binary model has a negative volume reward, pairwise geometry conflicts and a squared residual for selected weight plus slack minus capacity. The neal check samples all bits in this model. HiGHS instead solves a binary linear model with conflict and payload constraints and can certify the best volume for a fixed candidate pool.

Timing and inference

A timed method pays for pool or order construction, selection and validation. Results completed after the deadline are discarded. HiGHS gets the hybrid’s first pool and a solver time limit after setup. Confidence intervals use paired bootstrap resampling of cases after averaging seeds within each case.

Paired comparisons and attempt counts

The intervals below compare hybrid utilization minus First-Fit in percentage points. The nine-case five-second row uses three cases per family.

Limit	All	Geometry	Payload	Mixed
4 attempts	+2.66 [+0.95 to +4.51]	-0.62 [-2.26 to +1.09]	+0.00 [+0.00 to +0.00]	+8.60 [+6.33 to +10.71]
150 ms	+0.22 [-0.68 to +1.07]	-0.95 [-3.04 to +1.17]	+0.00 [+0.00 to +0.00]	+1.62 [+0.42 to +2.89]
500 ms	+0.80 [-0.02 to +1.59]	-0.26 [-1.89 to +1.44]	+0.00 [+0.00 to +0.00]	+2.65 [+1.56 to +3.58]
1.5 s	+0.64 [+0.06 to +1.29]	+0.12 [-1.04 to +1.61]	+0.00 [+0.00 to +0.00]	+1.81 [+0.89 to +2.72]
5 s subset	+0.79 [+0.00 to +2.04]	+1.81 [+0.00 to +5.42]	+0.00 [+0.00 to +0.00]	+0.57 [+0.00 to +1.08]

Limit	Cases	First-Fit	Hybrid	Mean attempts FF / hybrid
4 attempts	30	67.35%	70.01%	3.5 / 3.5
150 ms	30	69.17%	69.40%	255.2 / 2.6
500 ms	30	69.77%	70.57%	856.4 / 8.8
1.5 s	30	70.35%	70.99%	2,564.2 / 26.1
5 s subset	9	72.06%	72.85%	7,726.1 / 73.7

At a wall-clock deadline, faster attempts allow more restarts. At a fixed attempt cap, an attempt can cost more for one method. Report both views together.

The 150 ms, 500 ms and 1.5 s runs each use all 30 cases. Ten cases per family limit the precision of the family estimates. Timing may change with system load.

Mixed pools and candidate coverage

Time-limited HiGHS on mixed cases

Limit	HiGHS mean	Hybrid mean	On-time HiGHS	Proven on time
150 ms	26.21%	82.40%	30/30	4
500 ms	84.17%	85.23%	30/30	17
1.5 s	84.32%	86.10%	30/30	30
5 s subset	83.94%	86.41%	9/9	9

HiGHS builds one pool per trial. Its mean includes the empty fallback when no candidate result finishes on time. A certified optimum applies to that pool only. The hybrid can try multiple pools within a deadline.

Original mixed-pool diagnosis

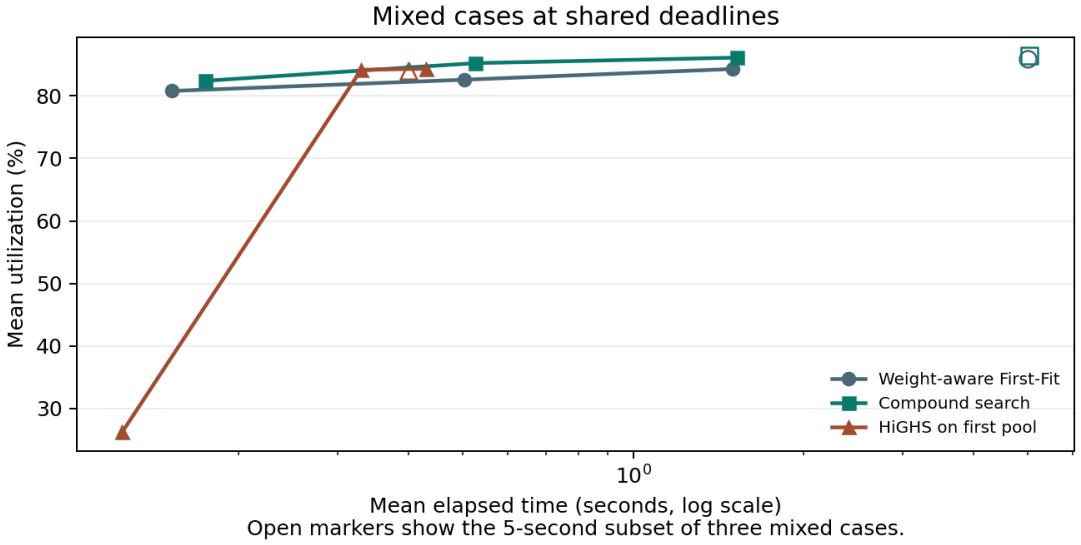
Across 30 recorded mixed pools, HiGHS proved 85.62% mean utilization versus 83.41% for the hybrid. The hybrid matched 10 of the 30 optima. The 2.21-point gap was available to better selection on those fixed pools. The 98.59% weight-and-volume bound ignores geometry.

Original geometry-pool diagnosis

Across 30 earlier geometry pools, candidate coverage accounted for 4.55 points of the gap to a full tiling. Search accounted for 0.21 points. In a three-case sensitivity check, varying corner priority raised one 20-box pool from 91.25% to 100%. This follow-up enables the change across all families.

Payload control

Knapsack followed by First-Fit matched the 28.02% payload upper bound on all ten payload cases. The same control averaged 70.26% on mixed cases. The payload family shows the benefit of choosing boxes by weight. The mixed family also requires the selected boxes to fit together.



Explicit QUBO and full support

Penalty sensitivity

Family	Original penalty	Best tested	Compound	Valid raw reads
Payload	25.83%	32.15%	32.15%	53/640
Mixed	53.54%	85.38%	86.01%	321/768

Values are mean best utilization after repair across three pools per family. The grid varied the squared slack penalty and used exact weight units or 1 kg units. Best tested uses the best setting for each pool and is an exploratory upper estimate. It does not measure how a frozen tuning rule generalizes.

The largest coefficient can dominate a box reward. This comparison changes that ratio and reports both raw validity and repaired volume. A weaker penalty can improve the search signal while making capacity violations cheaper.

Support rules

Original support-aware run	First-Fit	Hybrid	Gap (points)
Geometry	98.14%	91.33%	-6.81
Payload	27.71%	27.99%	+0.28
Mixed	78.03%	79.58%	+1.55

Checking support during search avoids losing a box when its base loses contact. The post-repair-only mode removes unsupported boxes and does not refill freed payload. This is a contact-area rule. Load-bearing strength and physical stability are separate engineering checks.

Reproduction and hardware appendix

Evidence and reproduction

The project README gives the Python setup. Run effort_benchmark.py --followup for the three-seed sweep. Add --fixed-only for reproducible attempt-capped scores. The effort_v4 results record attempts, time, solver statuses, paired intervals and returned placements. Timed scores can vary by machine.

Run explicit_qubo.py --tune with the optional neal dependency for the penalty sensitivity check. The test suite also verifies that the explicit QUBO minimized over slack matches the analytic penalty on a small fixture.

IBM Quantum fixtures

The earlier IBM ibm_fez run decoded the expected H0 pattern on 1,013 of 1,024 shots. The H1, H2 and H3 small packing fixtures hit their exact raw optimum in 31.00%, 24.96% and 1.54% of shots. These fixtures test circuit execution and decoding and are separate from the classical comparison.

Next experiment

Test on new mixed instances and hold the selected settings fixed. Separate coverage gains from selection gains by measuring the exact optimum on each recorded pool. Test geometry coverage with more than one corner-priority schedule. Treat physical support as an added constraint when using the results for realistic loading studies.

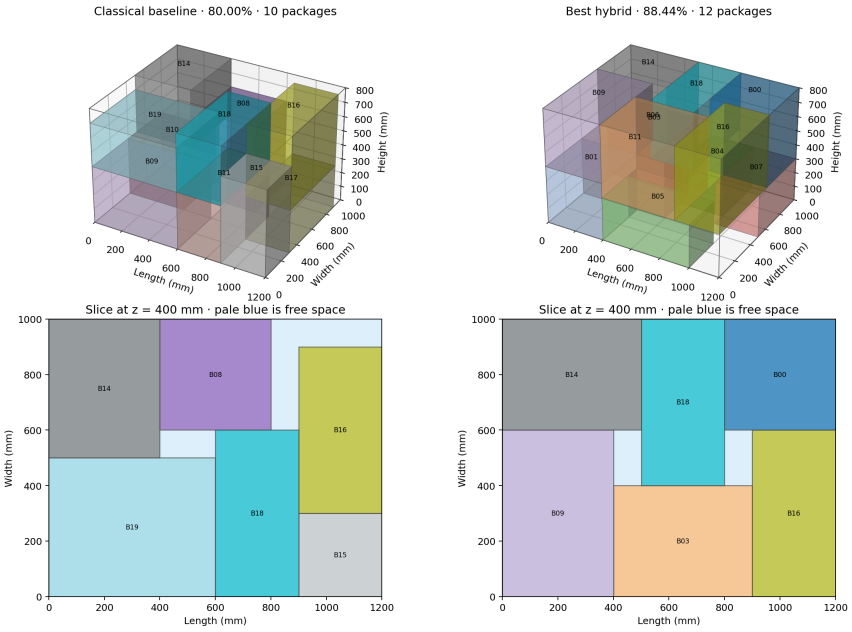


Figure 2. Earlier recorded mixed case 4302. First-Fit is at left and the hybrid is at right. Pale blue in the lower slice views shows free area.

Primary files include results/effort_v4/intervals.csv, summary.csv, packings.json and explicit_qubo_tuning.csv. Historical benchmark and hardware results remain in their original folders.